

Willis Electric Co., Ltd. v. Holiday Designs (Cambodia) Co. Ltd.

Public complaint analysis reports assertion of U.S. Patent Nos. 8,454,186 and 8,454,187 against pre-lit artificial trees, including the Maysville Pine and Forest Pine. The patents describe modular trunk sections carrying electrical connectors arranged so that electrical coupling can be made substantially independently of relative rotational orientation during assembly.

Independent preliminary research based on public information.

FILED	FORUM	DOCKET	MATTER
2026-08-18	E.D. Tex.	2:26-cv-00706	Patent Infringement

Prepared 2026-08-25. No attorney or expert contacted. Availability, interest, compensation, and conflicts not checked.

Public filing, likely recipient, and technical frame

CAPTION	Willis Electric Co., Ltd. v. Holiday Designs (Cambodia) Co. Ltd.
FORUM / DOCKET	E.D. Tex. / 2:26-cv-00706
FILED / TYPE	2026-08-18 / patent infringement
PUBLIC DOCKET	Open docket

VERIFIED PUBLIC RECORD

The public federal docket identifies Willis Electric Co., Ltd. as plaintiff and Holiday Designs (Cambodia) Co. Ltd. as defendant, verifies filing on August 18, 2026 in the Eastern District of Texas under docket 2:26-cv-00706, characterizes the case as patent infringement, and identifies Charles W. Goehringer, Jr. as counsel appearing for Willis and as the complaint filer.

LIKELY RESEARCH PURCHASER

Charles W. Goehringer, Jr. - Outside counsel appearing for Willis Electric; attorney at Germer PLLC. The exact public docket identifies Goehringer as counsel for Willis Electric and as the complaint filer. Selecting him as a likely purchaser is an analytical inference from that verified matter role; the public record reviewed does not identify who controls expert-retention decisions.

cwgoehringer@germer.com - publicly printed in the [linked professional source](#); not inferred.

TECHNICAL FRAME

Public complaint analysis reports assertion of U.S. Patent Nos. 8,454,186 and 8,454,187 against pre-lit artificial trees, including the Maysville Pine and Forest Pine. The patents describe modular trunk sections carrying electrical connectors arranged so that electrical coupling can be made substantially independently of relative rotational orientation during assembly.

Secondary-source limitation: The public docket and secondary complaint analysis verify the matter and reported patents, but the complaint PDF and accused specimens were not available in the reviewed record. The filed pleading, asserted claims, prosecution histories, accused-product versions, samples, drawings, and discovery must control any infringement or validity opinion. [Open public synopsis](#)

REPORTED ASSERTED PATENTS

[U.S. Patent No. 8,454,186](#) | [U.S. Patent No. 8,454,187](#)

Questions likely to require expert input

This issue map is a research plan, not a legal conclusion. The filed complaint, patent exhibits, intrinsic record, and discovery must control.

01 Does each accused tree contain the claimed modular trunk-and-connector arrangement?

An element-by-element inspection should identify trunk portions, first and second electrical connectors, conductors, mechanical engagement, and the relationship between tree-section assembly and electrical coupling.

Potential evidence: Complaint and claim charts; asserted claims and prosecution histories; accused Maysville Pine and Forest Pine specimens; assembly manuals; teardown photography; CAD and bills of materials

02 Is electrical coupling substantially independent of rotational orientation?

The central reported feature is coupling without requiring a single keyed angular position. Testing should measure the angular range, permitted orientations, insertion path, alignment features, and circumstances in which contact fails.

Potential evidence: Instrumented assembly over full rotation; continuity and resistance logging; connector geometry measurements; high-speed video; product instructions and engineering specifications

03 Which structures perform mechanical alignment and which perform electrical contact?

The connector carrier, trunk walls, guides, stops, contacts, springs, and conductors should be separated functionally to determine whether the accused architecture matches the claimed relationships rather than merely achieving a similar result.

Potential evidence: CT or section imaging; disassembly; dimensional metrology; drawings, tolerances, and supplier specifications; force-displacement testing

04 How reliable is the electrical path under repeated consumer assembly?

Contact resistance, wiping action, insertion force, wear, contamination, bending, torsion, and environmental exposure may reveal how the connector operates and whether product variants perform consistently.

Potential evidence: Cycle testing; four-wire resistance measurements; vibration and torsion tests; temperature/humidity exposure; microscopy; failure analysis

Questions likely to require expert input

05 Which claim terms require technical construction?

Potential disputes may concern trunk portion, electrical connector, electrical connection, rotational orientation, coaxial or concentric arrangement, releasable coupling, and the structural relationships stated in the asserted claims.

Potential evidence: Patents, prosecution histories, cited references, contemporaneous connector texts and standards, inventor records, and skilled-artisan declarations

06 Were the reported features present in every accused SKU and revision?

The Maysville Pine, Forest Pine, seasonal revisions, regional variants, suppliers, and replacement parts may differ. A reliable opinion needs version-specific mapping and date control.

Potential evidence: SKU/date-code matrix; archived manuals and marketing; engineering change orders; purchase and sales records; supplier drawings; comparison teardowns

07 What prior art disclosed rotation-tolerant electrical coupling in modular structures?

Validity work should search artificial trees and adjacent fields such as sectional poles, luminaires, furniture, coaxial contacts, slip-ring arrangements, and blind-mate connectors, while testing the claimed combination rather than isolated features.

Potential evidence: Patents, catalogs, manuals, standards, archived web pages, dated specimens, industry publications, and element-by-element charts

08 What noninfringing designs or alternative implementations are technically feasible?

Keyed connectors, separate cords, limited-angle contacts, plug-and-socket leads, or changed carrier geometry may affect the claim mapping. Any design-around analysis should preserve safety and consumer assembly performance.

Potential evidence: Alternative product specimens; controlled prototypes; electrical safety testing; CAD; tolerance studies; engineering and cost analysis

Candidates 1-2 for further diligence

Ranked for preliminary research priority based on subject-matter fit, relevant public work, testimony, and visible diligence burden. This is not a retention recommendation.

01 David O. Kazmer

Professor of Plastics Engineering, University of Massachusetts Lowell

TECHNICAL FIT

Strong fit for integrated plastic-product and process design, molded housings, dimensional tolerances, assembly, stress analysis, testing, and failure analysis. He could connect the asserted coaxial trunk-connector geometry to how molded tree sections are actually manufactured and assembled.

RELEVANT WORK

UMass Lowell identifies him as a named inventor on more than 20 patents and author of two books, ten book chapters, and more than 300 publications. His profile lists prior engineering and management roles at GE Research and Development, GE Plastics, and Synventive Molding Solutions.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, declaration, deposition, or trial-testimony source was located in the public sources reviewed.

POINTS FOR FURTHER DILIGENCE

His breadth is strongest for plastics manufacturing and integrated product-process design, not shed-specific building-code practice. University and prior industry relationships require conflict diligence. No public source reviewed establishes artificial-tree or coaxial trunk-connector work, so electrical-contact questions may require a complementary contacts specialist.

[Source 1](#) | [Source 2](#)

[Draft email](#) | David_Kazmer@uml.edu | [public email source](#)

02 Robert A. Malloy

Professor Emeritus and former Chair of Plastics Engineering, University of Massachusetts Lowell

TECHNICAL FIT

Strong fit for plastic part design, injection molding, mold design, thermoplastics, tolerancing, and failure-sensitive consumer-product design. Relevant to molded trunk shells, connector retention, ribs, guides, and rotation-independent mechanical interfaces.

RELEVANT WORK

UMass Lowell describes him as an active researcher and consultant, author of many publications and patents including a textbook on plastic part design, former chair of the Society of Plastics Engineers Injection Molding Division, SPE Fellow, and Plastics Hall of Fame member.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, declaration, deposition, or trial-testimony source was located in the public sources reviewed.

POINTS FOR FURTHER DILIGENCE

Emeritus status and current capacity require confirmation. His public profile does not establish knowledge of the specific Keter or Suncast products. No public source reviewed establishes artificial-tree or coaxial trunk-connector work, so electrical-contact questions may require a complementary contacts specialist.

[Source 1](#) | [Source 2](#)

[Draft email](#) | robert_malloy@uml.edu | [public email source](#)

Candidates 3-4 for further diligence

03 Tim A. Osswald

Professor of Mechanical Engineering and Co-Director, Polymer Engineering Center, University of Wisconsin-Madison

TECHNICAL FIT

Strong fit for polymer processing, injection molding, manufacturing simulation, material behavior, and design with polymers. Relevant to whether molded trunk and connector structures can achieve the claimed alignment, rigidity, and repeatable assembly.

RELEVANT WORK

UW-Madison identifies him as author of 12 polymer-engineering books and more than 300 technical papers, with expertise in polymer processing, composites, manufacturing, and design with polymers. His books include the Injection Molding Handbook.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, declaration, deposition, or trial-testimony source was located in the public sources reviewed.

POINTS FOR FURTHER DILIGENCE

His public record emphasizes polymer engineering broadly rather than folding mechanisms or outdoor-shed certification. Polymer Engineering Center industry relationships require conflict review. No public source reviewed establishes artificial-tree or coaxial trunk-connector work, so electrical-contact questions may require a complementary contacts specialist.

[Source 1](#) | [Source 2](#)

[Draft email](#) | tosswald@wisc.edu | [public email source](#)

04 Lih-Sheng Turng

Consolidated Papers Foundation Professor and Co-Director, Polymer Engineering Center, University of Wisconsin-Madison

TECHNICAL FIT

Strong fit for injection molding, computer-aided engineering, process control, dimensional stability, and design optimization. Relevant to manufacturing tolerances and variant analysis for nested trunk sections and embedded connector carriers.

RELEVANT WORK

UW-Madison reports more than 330 refereed journal papers, over 250 conference papers, 24 patents and applications, ten years developing injection-molding simulation software in industry, and extensive work on injection molding and process control.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, declaration, deposition, or trial-testimony source was located in the public sources reviewed.

POINTS FOR FURTHER DILIGENCE

His strongest fit is molding and simulation rather than bar-support kinematics or building-envelope performance. Extensive industrial sponsorships require conflict diligence. No public source reviewed establishes artificial-tree or coaxial trunk-connector work, so electrical-contact questions may require a complementary contacts specialist.

[Source 1](#) | [Source 2](#)

[Draft email](#) | turng@engr.wisc.edu | [public email source](#)

Candidates 5-6 for further diligence

05 Stephen P. Johnston

Professor of Plastics Engineering, University of Massachusetts Lowell

TECHNICAL FIT

Strong fit for plastic part and mold design, finite-element analysis, process control, and design for manufacturability. Relevant to mechanical retention, rib and wall geometry, connector support, and cyclic assembly testing.

RELEVANT WORK

UMass Lowell reports industrial experience with Moldflow, Bausch & Lomb, and Lord; more than 40 articles and proceedings; three patents; and teaching in product design, mold design, advanced mold design, and injection molding.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific expert report, declaration, deposition, or trial-testimony source was located in the public sources reviewed.

POINTS FOR FURTHER DILIGENCE

His public work is injection-molding centered, while some claims require blow-molding and tack-off analysis. Sponsored projects and industrial teaching relationships require conflict review. No public source reviewed establishes artificial-tree or coaxial trunk-connector work, so electrical-contact questions may require a complementary contacts specialist.

[Source 1](#) | [Source 2](#)

[Draft email](#) | Stephen_Johnston@uml.edu | [public email source](#)

06 Michael G. Pecht

Distinguished University Professor and George E. Dieter Professor of Mechanical Engineering; Founding Director, Center for Advanced Life Cycle Engineering, University of Maryland

TECHNICAL FIT

Exceptional fit for electronic-product reliability, electrical contacts and connectors, separable-contact design, qualification, accelerated testing, and failure analysis. His published connector work addresses contact geometry, normal force, wipe, corrosion, and contact resistance.

RELEVANT WORK

UMD identifies him as CALCE founding director and a specialist in product reliability, risk, qualification, and electronics systems. CALCE publishes his work on gold-plated electric contacts and a reference covering contacts, connectors, and IC sockets.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific public expert report, declaration, deposition, or trial-testimony record for Michael G. Pecht was located in the bounded public-source review. This is not proof that the candidate has never served in litigation; PACER, paid expert databases, retention records, and conflict systems were not searched.

POINTS FOR FURTHER DILIGENCE

His core expertise is electronics reliability rather than decorative artificial-tree construction. CALCE has extensive industry relationships that require conflict diligence.

[Source 1](#) | [Source 2](#) | [Source 3](#)

[Draft email](#) | pecht@umd.edu | [public email source](#)

Candidates 7-8 for further diligence

07 Michael H. Azarian

Research Scientist, Center for Advanced Life Cycle Engineering, University of Maryland

TECHNICAL FIT

Direct fit for electrical contacts, passive components, failure analysis, prognostics, and product reliability. Relevant to whether repeated trunk rotation, insertion, contamination, wear, or misalignment changes electrical continuity and claimed connector operation.

RELEVANT WORK

CALCE publicly lists his focus as failure analysis, prognostics and health management, electrical contacts, passive components, and counterfeit detection and mitigation.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific public expert report, declaration, deposition, or trial-testimony record for Michael H. Azarian was located in the bounded public-source review. This is not proof that the candidate has never served in litigation; PACER, paid expert databases, retention records, and conflict systems were not searched.

POINTS FOR FURTHER DILIGENCE

Public materials establish contacts and reliability expertise but not artificial-tree product design or patent-testimony experience. Current industry and consortium relationships require conflict diligence.

[Source 1](#) | [Source 2](#)

[Draft email](#) | mazarian@umd.edu | [public email source](#)

08 Diganta Das

Associate Research Scientist, Mechanical Engineering and Center for Advanced Life Cycle Engineering, University of Maryland

TECHNICAL FIT

Strong fit for electronic-part reliability, environmental and operational ratings, component selection, defects, and accelerated screening. Relevant to connector durability, manufacturing variation, contamination, and reliability across product versions.

RELEVANT WORK

UMD reports more than 75 publications, IEEE reliability and standards roles, and research in electronic-component reliability. CALCE identifies connector defects and stress-screening work and lists him as a contact for its connector reference.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific public expert report, declaration, deposition, or trial-testimony record for Diganta Das was located in the bounded public-source review. This is not proof that the candidate has never served in litigation; PACER, paid expert databases, retention records, and conflict systems were not searched.

POINTS FOR FURTHER DILIGENCE

His work is broader component reliability rather than the claimed mechanical rotation geometry. Public sources do not establish artificial-tree-specific work.

[Source 1](#) | [Source 2](#) | [Source 3](#)

[Draft email](#) | diganta@umd.edu | [public email source](#)

Candidates 9-10 for further diligence

09 Michael Osterman

Research Scientist and Director of CALCE Electronic Products and Systems Consortium Operations, University of Maryland

TECHNICAL FIT

Strong fit for testing and simulation-based failure assessment of electronic products. Relevant to developing repeatable electrical-continuity, insertion-cycle, torsional, vibration, environmental, and teardown protocols for the accused tree connectors.

RELEVANT WORK

CALCE publicly identifies him with testing and simulation-based failure assessment and publishes his professional contact through its team directory.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific public expert report, declaration, deposition, or trial-testimony record for Michael Osterman was located in the bounded public-source review. This is not proof that the candidate has never served in litigation; PACER, paid expert databases, retention records, and conflict systems were not searched.

POINTS FOR FURTHER DILIGENCE

The public profile is more concise than those of the senior faculty and does not establish tree-specific, claim-construction, or testimony experience.

[Source 1](#)

[Draft email](#) | osterman@umd.edu | [public email source](#)

10 Abhijit Dasgupta

Professor Emeritus of Mechanical Engineering, University of Maryland, affiliated with CALCE

TECHNICAL FIT

Strong fit for interconnect reliability, accelerated testing, mechanics, and failure mechanisms. Relevant to connector carrier stresses, repeated assembly, contact support, and distinguishing design limitations from manufacturing defects.

RELEVANT WORK

CALCE materials publicly identify him with interconnect reliability and accelerated testing; his research record addresses mechanics-of-failure approaches for electronic packaging and interconnects.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific public expert report, declaration, deposition, or trial-testimony record for Abhijit Dasgupta was located in the bounded public-source review. This is not proof that the candidate has never served in litigation; PACER, paid expert databases, retention records, and conflict systems were not searched.

POINTS FOR FURTHER DILIGENCE

Emeritus status and current professional capacity require confirmation. His principal public work concerns electronic packaging rather than consumer artificial trees.

[Source 1](#) | [Source 2](#)

[Draft email](#) | dasgupta@umd.edu | [public email source](#)

Preliminary candidates; no conflicts or availability determination

- Run party, affiliate, counsel, inventor, funder, and technology conflicts.
- Confirm testimony history, exclusions, compensation, publications, patents, and deposition performance.
- Investigate licensing, grants, consulting, commercial interests, and prior technical positions.
- Confirm role fit, availability, and interest only after counsel authorizes outreach.

ONGOING MATTER-SPECIFIC RESEARCH

Flint Witness Research works in the post-filing window, turning pleadings and public technical records into issue maps, researched candidate slates, and diligence starting points as the case develops.

[Schedule a 15-minute conversation](#)

CORE SOURCES

Public docket: <https://dockets.justia.com/docket/texas/txedce/2%3A2026cv00706/248476>

Public professional email source: <https://cases.iustia.com/federal/district-courts/texas/txedce/9%3A2010cv00035/121662/22/0.pdf>

Public technical context source: <https://ai-lab.exparte.com/case/dct/txed/2%3A26-cv-00706/doc/analysis/1>

LIMITATIONS

The public docket and secondary complaint analysis verify the matter and reported patents, but the complaint PDF and accused specimens were not available in the reviewed record. The filed pleading, asserted claims, prosecution histories, accused-product versions, samples, drawings, and discovery must control any infringement or validity opinion.

No attorney or expert was contacted. No availability, interest, mailbox delivery, or conflict clearance is claimed.